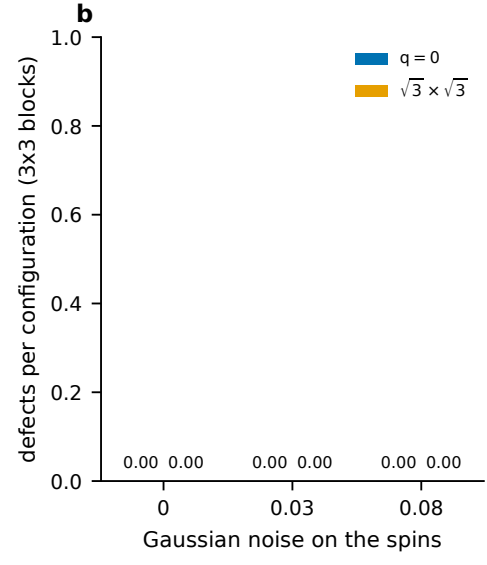
a **classification in the binary dihedral group $2D_3$**

none	trivial: frame returns to itself
Z_2	returns up to -1 in the double cover
C_3	triad relabelled cyclically (1/3 vortex)
C_2	plane normal reversed (1/2 dislocation)

Frame per block: ν = min. eigenvector of Q, triad phase arg $\langle e^{3i\phi} \rangle / 3$; candidates ±, 3 cyclic phases; lift along the loop of 4 blocks.